Lab2: Tuple & Set

COCS-221: Computer Science B

Objective

By the end of this lab, you will be able to

- Learn how to create and modify tuples and sets in Python.
- Perform basic set operations like union, intersection, and difference.
- Use tuple methods to search for data.
- Work with lists and sets, converting lists to sets and managing set items.
- Learn how to save and load data to/from a file.
- Practice handling user input and manipulating data in Python.

Exercise

Exercise 1:

- Write a Python program to create a tuple with different data types.
- Write a Python program to add "please add me" text and your birthday to the tuple that you just created.
- Write a Python program to check whether your birthday that you just added exists within a tuple.
- Write a Python program to find the length of a tuple.

Exercise 2:

- Create a tuple containing at least five numbers and print the first and last element.
- Create two sets of numbers (example: {1, 2, 3, 4} and {3, 4, 5, 6}) and:
  - Perform union(), intersection(), and difference().
  - Print the results of each operation.

Exercise 3:

- Write a program that ask the user to input three values: name, age, and city, and store these in a tuple.
- Use tuple methods such as count() and index() to search for specific elements:

- Verify that your result should be:

```
Enter your name: Mat Nab
Enter your age: 30
Enter your city: Phnom Penh
Your tuple: ('Mat Nab', 30, 'Phnom Penh')
Count of 'Vanda' in the tuple: 0
Index of 'Phnom Penh' in the tuple: 2
```

Exercise 4:

- Take a list of numbers [1, 2, 3, 4, 4, 5, 6, 6, 7, 8], convert it to a set and display the set.
- Write a program to add 10, 20, 30 to a set
- Write a Python program to remove three last items from the following set at once.
- Checking if 6 is a member of a set
- Finding the length of a set

Challenging Exercise1

- Create a tuple with some student names.
- Convert the tuple to a set to remove any duplicates.
- Search for a name in the set.
- Add or remove names from the set based on user input.
- Save the set of names to a file.
- Load the names from the file and display them.
- Verify that your result should be:

```
Set after removing duplicates: {'Reaksa', 'Chan Panha', 'Mani', 'Amara', 'Sreytim', 'Laksmey', 'Khayhour'}
Enter a name to search: Mat Nab
Mat Nab is not in the set.
Do you want to add or remove a name? (add/remove): add
Enter a name to add: Mat Nab
Updated set: {'Reaksa', 'Chan Panha', 'Mani', 'Amara', 'Sreytim', 'Laksmey', 'Khayhour', 'Mat Nab'}
Names saved to file.
Names loaded from file: {'Reaksa', 'Chan Panha', 'Mani', 'Sreytim', 'Amara', 'Laksmey', 'Khayhour', 'Mat Nab'}
```

Challenging Exercise2

- Write a function dice_roll(guess = (1, 3), n = 100): that takes a tuple of guessing values on the dice 'guess' and the number of rolls n. The function should return the proportion of correctly guessed and the computational time.
- If you guess (1, 6) and n=100000, what is the expected result? Print the result and the running time.

End

Great job! Keep practicing, and it will get easier. Stay motivated, you're doing great!